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Teorema de Euler-Lagrange

Teorema 1 (Euler-Lagrange). *Sea* $f \in \mathcal{C}^2([a,b]\times\mathbb{R}\times\mathbb{R})$ *e* $y \in \mathcal{C}^2([a,b])$ *mínimo o máximo de la integral* $J[y] = \displaystyle\int_a^b f(x, y(x), y'(x))\,\mathrm{d}x$*, con* $y(a) = y_a \wedge y(b) = y_b$

$$\implies \forall x \in [a,b] : f_y(x, y(x), y'(x)) - \frac{\mathrm{d}}{\mathrm{d}x}\left(f_{y'}(x, y(x), y'(x))\right) = 0.$$

***Demostración*:** Tomamos $\varphi \in \mathcal{C}^2([a,b])$ con $\varphi(a) = 0$ y $\varphi(b) = 0$ y consideramos

$$J[y+\varepsilon\varphi] = \int_a^b f(x, y(x)+\varepsilon\varphi(x), y'(x)+\varepsilon\varphi'(x))\,\mathrm{d}x =: \overline{J}(\varepsilon), \quad \text{para } \varepsilon \in \mathbb{R}.$$

La función $\overline{J}\colon \varepsilon \mapsto \overline{J}(\varepsilon)$ tiene un máximo o mínimo en $\varepsilon = 0$ por hipótesis, por tanto su derivada se anula en $\varepsilon = 0$:

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \int_a^b f(x, y(x)+\varepsilon\varphi(x), y'(x)+\varepsilon\varphi'(x))\,\mathrm{d}x \\ &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} f(x, y+\varepsilon\varphi, y'+\varepsilon\varphi')\,\mathrm{d}x = \int_a^b (f_y\varphi + f_{y'}\varphi')\,\mathrm{d}x \end{aligned}$$

donde $f_y = \frac{\partial f}{\partial y}$ y $f_{y'} = \frac{\partial f}{\partial y'}$. Ahora bien, como $\displaystyle\int_a^b f_{y'}\varphi'\,\mathrm{d}x = f_{y'}\varphi\big|_a^b - \int_a^b \frac{\mathrm{d}}{\mathrm{d}x}(f_{y'})\,\varphi\,\mathrm{d}x$

$$\implies \int_a^b f_y\varphi\,\mathrm{d}x + \cancelto{0}{f_{y'}\varphi\big|_a^b} = \int_a^b \frac{\mathrm{d}}{\mathrm{d}x}(f_{y'})\,\varphi\,\mathrm{d}x \implies \int_a^b \left(f_y - \frac{\mathrm{d}}{\mathrm{d}x}(f_{y'})\right)\varphi\,\mathrm{d}x = 0.$$

Por tanto, por el `lem-dubois-reymond/`**??**, $\forall x \in [a,b] : f_y - \dfrac{\mathrm{d}}{\mathrm{d}x}(f_{y'}) = 0.$ ∎

Observación 1. Como $\frac{\mathrm{d}}{\mathrm{d}x}(f_{y'}) = f_{xy'} + y' f_{yy'} + y'' f_{y'y'}$, si $f_{y'y'} \neq 0$, podemos despejar y'':

$$y'' = \frac{f_y - f_{xy'} - y' f_{yy'}}{f_{y'y'}}.$$

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